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SRR — EOSAT-1 Multispectral Imager

System Requirements Review

SpaceCDF

2026

Faculty of Engineering • School of Engineering Design and Teaching Innovation (SED TI)

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Classification: Internal • Demo

Document Information

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1.0	2026-05-12	SpaceCDF	Initial SRR issue generated from converged design loop.

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This document was produced with the assistance of generative AI as part of the SpaceCDF Concurrent Design Facility workflow. The design-loop convergence, agent rationales, embedded figures and document rendering are generated by the SpaceCDF backend (Python · matplotlib · python-docx) guided by ECSS, NASA SEH and SMAD-4 references. Editorial framing, worked examples and pedagogical commentary remain owned by the SpaceCDF teaching team.

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Acronyms & Abbreviations

Acronym	Definition
AIT	Assembly, Integration and Test
AOCS	Attitude and Orbit Control Subsystem

BoL	Beginning of Life
CDF	Concurrent Design Facility
CDR	Critical Design Review
ConOps	Concept of Operations
DDT&E	Design, Development, Test and Evaluation
EPS	Electrical Power Subsystem
EoL	End of Life
FMECA	Failure Mode, Effects and Criticality Analysis
GS	Ground Station
IADC	Inter-Agency Space Debris Coordination Committee
ITU	International Telecommunication Union
MLI	Multi-Layer Insulation
MRD	Mission Requirements Document
MTBF	Mean Time Between Failures
OBDAH	On-Board Data Handling
PDR	Preliminary Design Review
RPN	Risk Priority Number
RW	Reaction Wheel
SA	Solar Array
SCDF	SpaceCDF Concurrent Design Facility
SRR	System Requirements Review
SSO	Sun-Synchronous Orbit
TCS	Thermal Control Subsystem
TRL	Technology Readiness Level
TT&C	Telemetry, Tracking and Command
VP	Verification Plan
WBS	Work Breakdown Structure

Applicable Documents

ID	Reference	Title
AD-01	ECSS-M-ST-10C Rev.1	Space project management
AD-02	ECSS-E-ST-10C Rev.1	System engineering general requirements
AD-03	ECSS-E-ST-10-02C	Verification requirements

AD-04	ECSS-E-ST-10-06C	Technical requirements specification
AD-05	ECSS-Q-ST-30C	Dependability
AD-06	ECSS-Q-ST-40C Rev.1	Safety

Reference Documents

ID	Reference	Title
RD-01	NASA SP-2016-6105	NASA Systems Engineering Handbook Rev. 2
RD-02	NPR 7123.1D	NASA Systems Engineering Processes and Requirements
RD-03	SMAD-4	Space Mission Analysis and Design (4th ed.)
RD-04	ECSS-U-AS-10C Rev.2	Space debris mitigation requirements
RD-05	ISO 24113:2023	Space debris mitigation requirements
RD-06	ITU-R P.618-13	Propagation data for Earth-space links
RD-07	ITU-R P.676-13	Attenuation by atmospheric gases

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1. Scope

1.1 Purpose

This document is the System Requirements Review for the EOSAT-1 Multispectral Imager. It captures the converged engineering baseline produced by the SpaceCDF AI Concurrent Design Facility and provides the evidence required for the SRR review gate per ECSS-M-ST-10C Rev.1.

1.2 Applicability

The contents are applicable to all engineering disciplines, project management, product assurance and operations. Subsystem-level data items inherit applicability from the parent applicable documents listed above.

1.3 Conventions

All units are SI unless otherwise stated. Margins are quoted relative to the nominal value (i.e. with-margin = nominal $\times$ (1 + margin%)). Status colours follow the SpaceCDF / ECSS convention: green = compliant, amber = marginal, red = non-compliant or exceeded.

2. Mission Overview

Mission 'EOSAT-1 Multispectral Imager' is a Earth Observation mission carried out by 1 nano-class spacecraft over a design life of 3.0 years. Top-level mission targets are 12.0 kg launch mass and 5.0 MEUR total programme cost.

Mission attribute	Value
Mission name	EOSAT-1 Multispectral Imager
Mission type	Earth Observation
Spacecraft class	nano
Number of spacecraft	1
Design lifetime (yr)	3.00
Reliability target	0.90
Cost target (MEUR)	5.00
Mass target (kg)	12.00
Ground stations	KSAT Svalbard, KSAT Troll

Table 1 — Top-level mission descriptor.

2.1 Payload complement

Name	Type	Mass (kg)	Power (W)	Rate (Mbps)	Pointing (°)	Duty
Multispectral Imager	optical_imager	1.50	20.0	100.0	0.100	25%

Table 2 — Payload complement.

3. Concept of Operations & Orbit

Parameter	Value
Orbit type	SSO
Altitude (km)	450
Inclination (deg)	97.40
Period (min)	93.4
Orbits per day	15.41
Velocity (m/s)	7644.4
Eclipse fraction	0.384
Footprint radius (km)	2327
Contact time / day (s)	2055
Mission duration (yr)	3.0

Table 3 — Orbital parameters.

Earth shadow

SSO · 450 km · $i=97.4^\circ$ · eclipse fraction 0.38

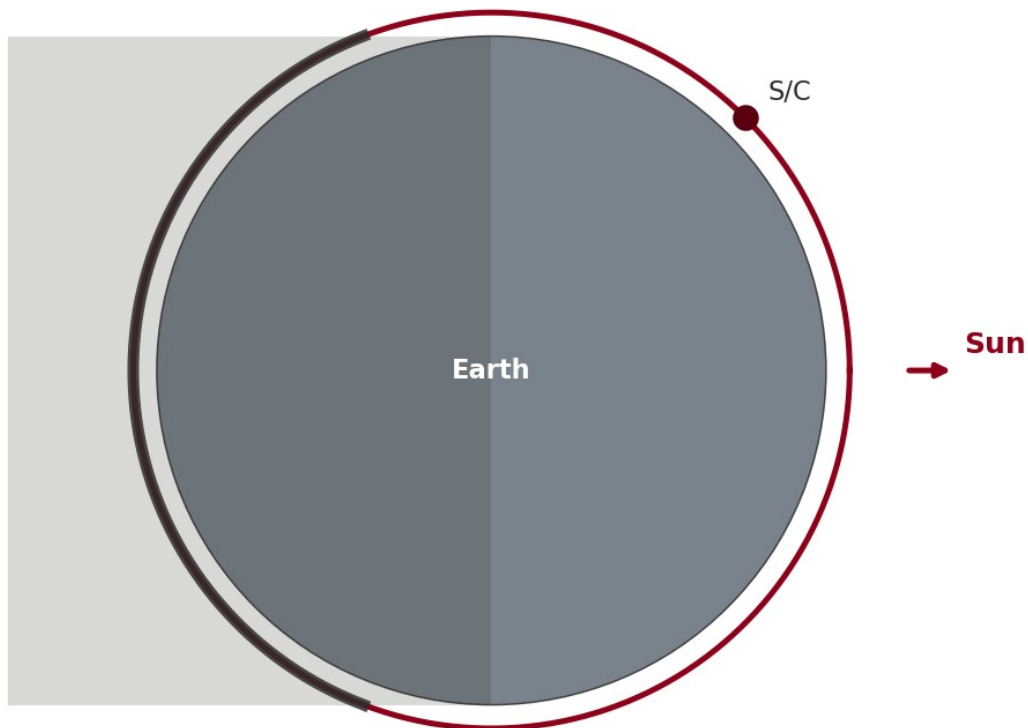


Figure 1 — SSO orbit geometry with shaded eclipse arc.

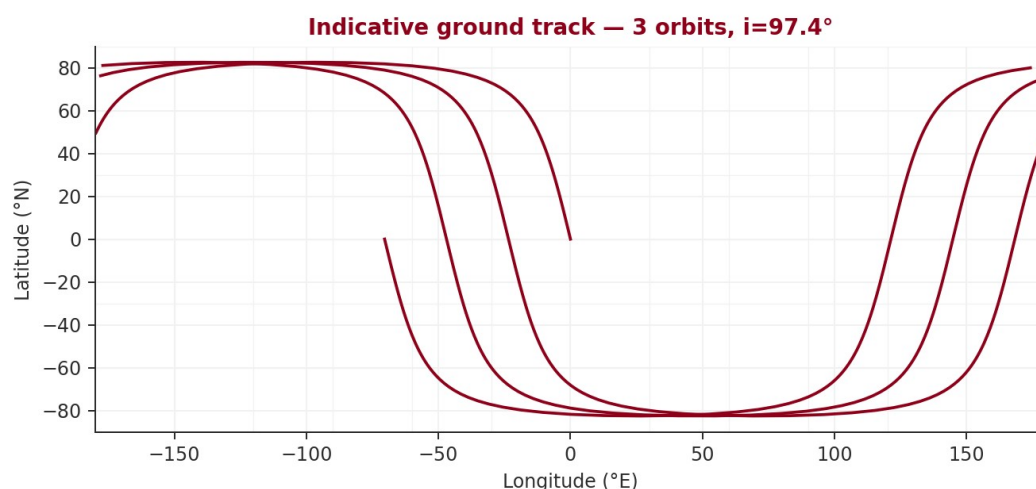


Figure 2 — Indicative ground track over three consecutive orbits.

Rationale

Orbit chosen as SSO at 450 km altitude and inclination 97.40° . The Keplerian period is 93.4 min (15.4 orbits / day) with an eclipse fraction of 0.38 (cylindrical shadow). A high-latitude ground station near 78° provides ≈ 34.2 min of contact / day.

Assumptions

- Cylindrical Earth-shadow eclipse model (no penumbra).
- Ground-station latitude assumed 78° (Svalbard-class).
- Station-keeping ΔV from JR-1971 drag model over 3-yr life.

3.1 ΔV breakdown

Component	ΔV (m/s)	Rationale
Station-keeping	4.6	Drag makeup over 3 yr design life.
Collision avoidance	3.0	Allowance of ~ 1 m/s per year for collision-avoidance burns.
Deorbit / EOL	71.7	Lower perigee for ≤ 25 -yr re-entry (IADC compliance).

Table 4 — ΔV budget breakdown.

4. System Architecture

The spacecraft system architecture follows the canonical CubeSat-class topology with a payload routing data through the OBDH to the TT&C subsystem, while the EPS distributes regulated power to all platform subsystems. AOCS drives propulsion (when fitted) and the thermal subsystem couples conductively to the structure.

System architecture — subsystem block diagram

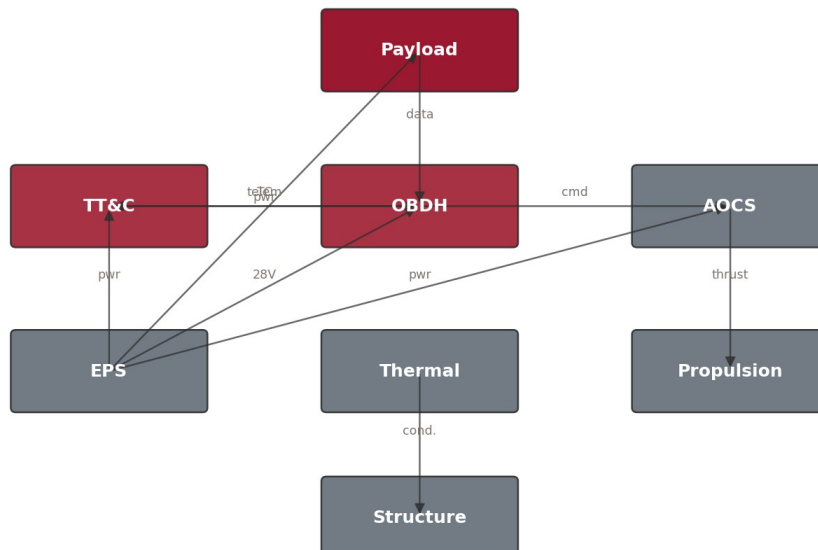


Figure 3 — Spacecraft system architecture — block diagram.

5. Payload

5.1 Multispectral Imager

4-band ocean colour imager (443, 560, 665, 865 nm)

Attribute	Value
Type	optical_imager
Mass (kg)	1.50
Power, mean (W)	20.0
Power, peak (W)	45.0
Data rate (Mbps)	100.00
Data volume / day (GB)	2.00
Pointing accuracy (°)	0.100
Duty cycle (%)	25

6. Electrical Power Subsystem

Rationale

Electrical power system sized for a worst-case sunlight load of 31.0 W and worst-case eclipse load of 36.0 W. Solar array delivers 76.5 W BoL / 70.9 W EoL after 3.0 yr of degradation, covering 0.216 m² of 29.5%-efficient cells. Battery capacity 71.7 Wh chosen for 30.0% depth-of-discharge over the eclipse duration.

Assumptions

- Solar cell efficiency 29.5% (class-appropriate triple-junction GaAs / KB-resolved cell).
- Specific power 80 W/kg.
- Mode profile estimated from payload duty cycles.
- Heater allocation 10.0 W during eclipse.
- Battery DoD limit 30% (ECSS Li-ion practice).

Parameter	Value	Unit	Margin
Battery Capacity	71.7	Wh	20%
Battery DOD	30	%	20%
Battery Mass	0.48	kg	20%
EPS Cost	12	kEUR	20%
EPS Total Mass	1.57	kg	20%
Solar Array Area	0.216	m ²	20%
Solar Array Mass	0.89	kg	20%
SA Power BOL	76.5	W	20%
SA Power EOL	70.9	W	20%
Total Power (Eclipse)	36	W	20%
Total Power (Sunlight)	31	W	20%

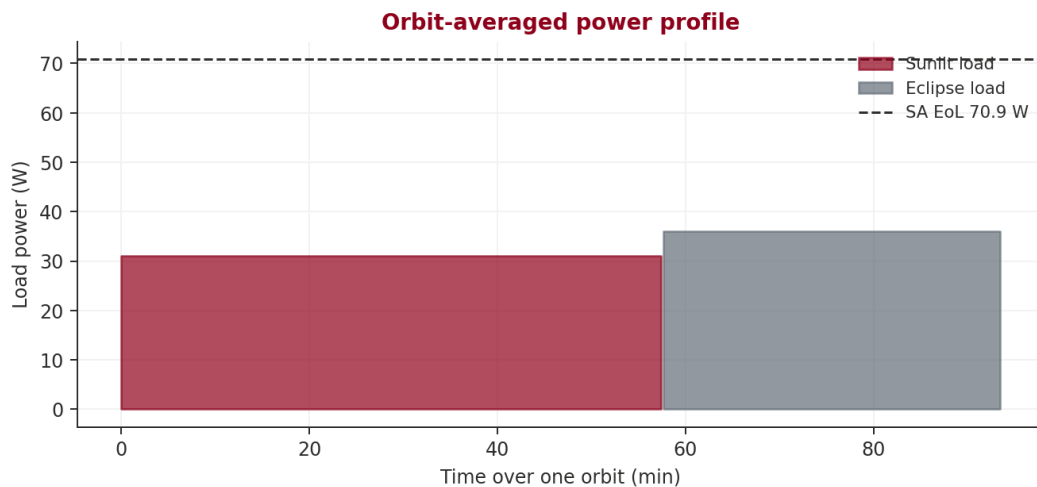


Figure 4 — Orbit-averaged power profile (sunlit / eclipse, SA EoL line).

6.2 Battery

Parameter	Value
Capacity (Wh)	71.7
Depth of discharge (%)	30.0
Mass (kg)	0.48
Chemistry	Li-ion (assumed)

7. Thermal Control Subsystem

Rationale

Thermal control sized for a hot-case internal dissipation of 31.0 W on a 0.22 m² external surface (cubesat). The radiator area is 0.113 m² ($\epsilon=0.85$, $\alpha=0.15$) and rejects to space at $T_{\text{max}}=50$ °C. In the cold case (eclipse, 30 % standby load) heaters supply 0.0 W to hold the platform above $T_{\text{min}}=-20$ °C.

Assumptions

- Stefan-Boltzmann steady-state energy balance, no transient analysis.
- Solar flux 1361 W/m², Earth IR 237 W/m², albedo 0.30.
- Radiator: $\epsilon=0.85$, $\alpha=0.15$ (white paint / OSR).
- MLI effective emissivity 0.01 (15-layer blanket).
- Platform allowable range -20 °C to +50 °C; payload limits set separately.

Warnings & open issues

- Radiator area 0.11m² exceeds 50% of S/C area — consider active cooling

Parameter	Value	Unit	Margin
Eclipse Heater Power	0	W	20%
Radiator Area	0.113	m ²	20%
Radiator Mass	0	kg	20%
TCS Cost	5	kEUR	20%
TCS Total Mass	0.35	kg	20%

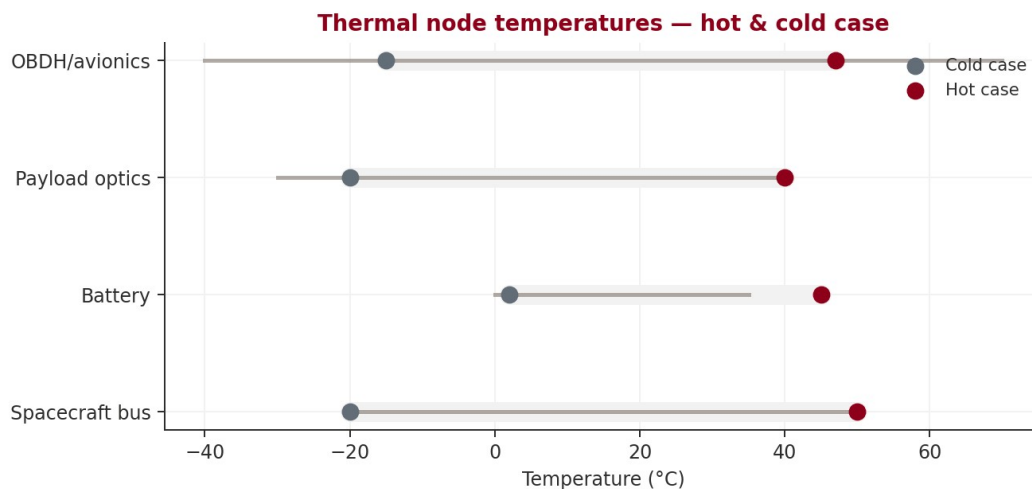


Figure 5 — Hot and cold case temperatures across the principal thermal nodes.

Node	Cold (°C)	Hot (°C)	Min limit	Max limit
Spacecraft bus	-20.0	50.0	-20.0	50.0
Battery	2.0	45.0	0.0	35.0
Payload optics	-20.0	40.0	-30.0	40.0

OBDH/avionics	-15.0	47.0	-40.0	70.0
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7.1 Surface treatments

Surface	α	ϵ	Area (m ²)
Radiator (OSR)	0.15	0.85	0.113
MLI (15-layer)	0.10	0.01	0.107

8. AOCS

Rationale

AOCS sized for a pointing requirement of 0.100° around a earth central body. Disturbance torque budget is dominated by gravity gradient (9.94e-09 Nm), with solar pressure (6.13e-09), aerodynamic (2.20e-08) and magnetic (2.44e-06) contributions over one orbit summing to 2.48e-06 Nm. Reaction wheel momentum capacity 0.003 Nms provides one-orbit absorption before desaturation.

Assumptions

- Spacecraft maximum dimension 0.30 m (class-typical).
- Wet area assumed 0.09 m² (cubical bus).
- Worst-case attitude offsets: 5° from nadir for gravity gradient; 0.05 m solar CoP offset; 0.05 m aero CoP offset.
- Magnetorquer dipole desaturation effective at mid-latitudes (Earth orbits only).

Parameter	Value	Unit	Margin
AOCS Cost	204	keUR	20%
AOCS Mass	2.04	kg	20%
Pointing Accuracy	0.1	deg	20%
AOCS Power	15	W	20%
Aerodynamic Torque	0	Nm	20%
Gravity Gradient Torque	0	Nm	20%
Magnetic Torque	2e-06	Nm	20%
Solar Pressure Torque	0	Nm	20%
Total Disturbance Torque	2e-06	Nm	20%
Wheel Momentum	0.003	Nms	20%

8.1 Disturbance torque budget

Source	Torque (Nm)
Gravity gradient	9.94e-09
Solar pressure	6.13e-09
Aerodynamic	2.20e-08

Magnetic	2.44e-06
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8.2 Pointing error budget

Contributor	1 σ (arcsec)
Sensor noise (star tracker)	10.0
Actuator quantisation (RW)	30.0
Disturbance reject. residual	60.0
Alignment	36.0
Achieved (RSS)	76.8

9. Propulsion

Rationale

No propulsion required — the spacecraft is a nano class at 450 km altitude, where natural atmospheric drag completes deorbit within the IADC 25-year window without active manoeuvres.

Assumptions

- Natural-decay regime (low-LEO nano/micro-class).
- No collision-avoidance budget; mission-dependent JCA on-orbit.

Parameter	Value	Unit	Margin
Propulsion Cost	0	kEUR	20%
Total Delta-V	0	m/s	20%
Specific Impulse	0	s	20%
Propellant Mass	0	kg	20%
Total Impulse	0	Ns	20%
Propulsion System Mass	0	kg	20%
Propulsion Type	none		20%

10. Telemetry, Tracking & Command

Rationale

Communications architecture is a dual link: an always-on UHF TT&C channel for command and housekeeping (margin 29.4 dB) and a higher-rate X payload downlink at 8.2 GHz (margin 7.7 dB). EIRP is 8.0 dBW; free-space loss is 176.5 dB at the worst-case slant range of 1944 km.

Assumptions

- 5° minimum elevation; ITU-R P.676/P.618 atmospheric model.
- GS antenna 13.0 m, efficiency 0.55, system noise temperature 200 K.

- Modulation coding rate 0.5, protocol overhead 10%, implementation loss 2 dB.
- Required Eb/N0 10 dB for BER 10^{-6} at the assumed modulation.

Parameter	Value	Unit	Margin
Data Downlinked/Day	2.4	GB	20%
Downlink Margin (worst)	7.7	dB	20%
Downlink Data Rate	9.34e+06	bps	20%
Payload DL Margin	7.7	dB	20%
TTC Cost	32	KEUR	20%
TTC Margin	29.4	dB	20%
TTC Mass	0.42	kg	10%
TTC Power	11	W	20%
Uplink Margin	39.4	dB	20%

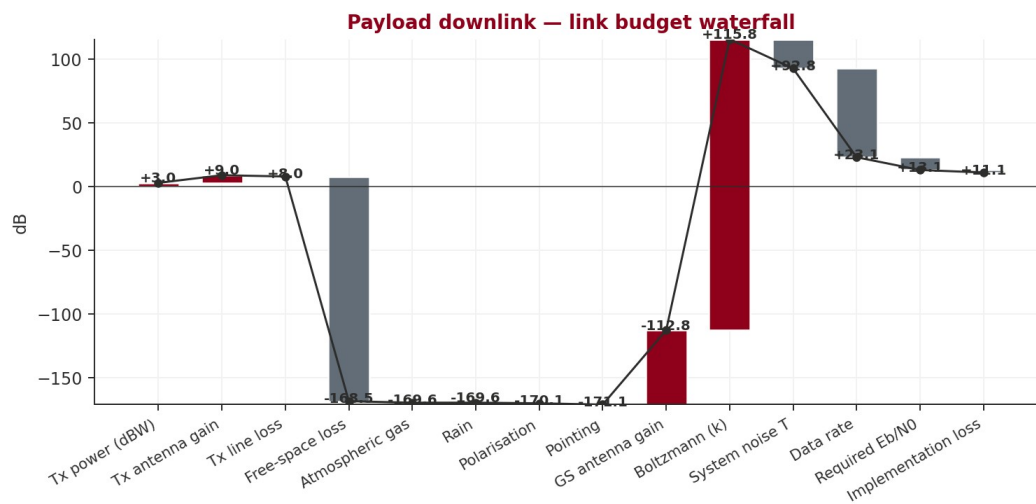


Figure 6 — Payload downlink link budget waterfall (positive = gain, negative = loss).

Term	Δ (dB)
Tx power (dBW)	+3.01
Tx antenna gain	+6.00
Tx line loss	-1.00
Free-space loss	-176.50
Atmospheric gas	-1.14
Rain	-0.00
Polarisation	-0.50
Pointing	-1.00
GS antenna gain	+58.37
Boltzmann (k)	+228.60
System noise T	-23.01

Data rate	-69.71
Required Eb/N0	-10.00
Implementation loss	-2.00

10.1 Link summary

Parameter	Value
Band	X
Frequency (GHz)	8.200
S/C Tx power (W)	2.0
S/C antenna gain (dBi)	6.0
GS antenna Ø (m)	13.00
GS antenna gain (dBi)	58.4
Slant range (km)	1944
User data rate (bps)	9,344,907
Data downlinked / day (GB)	2.40
Downlink margin (dB)	7.7
TT&C margin (dB)	29.4
Uplink margin (dB)	39.4

11. Data Handling

Rationale

Data architecture sized for 2.0 GB/day generated (2.4 GB/day downlinked). Onboard storage 4.0 GB provides 2-orbit redundancy. OBDH mass 0.16 kg from class factor × storage demand.

Assumptions

- 2:1 lossless compression when payload data rate is used as input (explicit per-day volume overrides this).
- Storage = 2 × daily generation (operational buffer).
- OBDH mass scales with class baseline × (1 + storage/100 GB).

Parameter	Value	Unit	Margin
Data Downlinked/Day	2.4	GB	20%
Data Generated/Day	2	GB	20%
OBDH Cost	16	KEUR	20%
OBDH Mass	0.16	kg	10%
Storage Required	4	GB	20%

12. Structure & Mechanisms

Rationale

Structure mass is taken from the heritage class-fraction database with Phase-A margin of 20%. Detailed FEM, modal analysis, and mechanism qualification are deferred to PDR/CDR.

Parameter	Value	Unit	Margin
Structure Cost	8	kEUR	20%
Structure Fraction	0.141		20%
Structure Mass	0.99	kg	20%

13. System Budgets

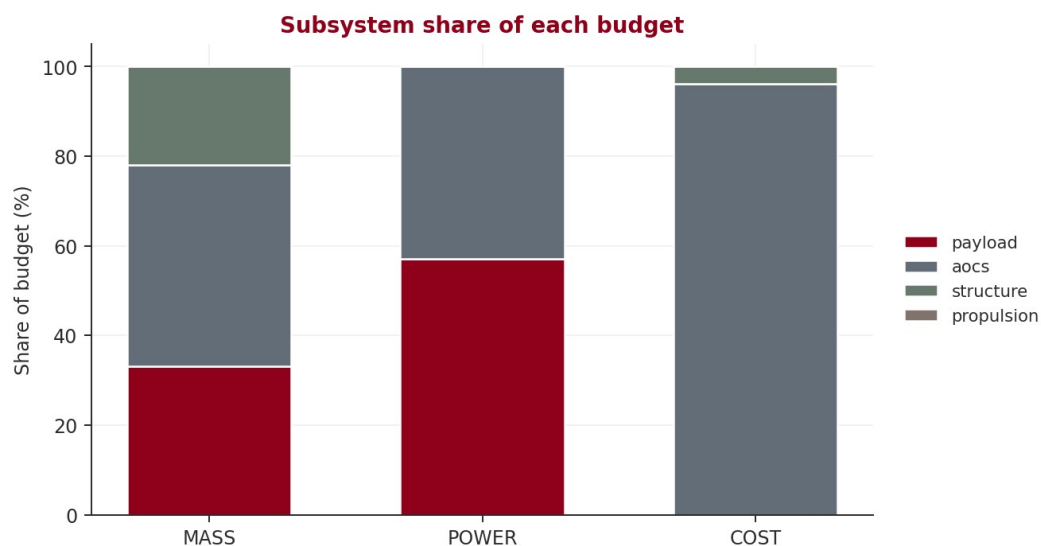


Figure 7 — Normalised share of each system budget by subsystem.

13.1 Mass budget

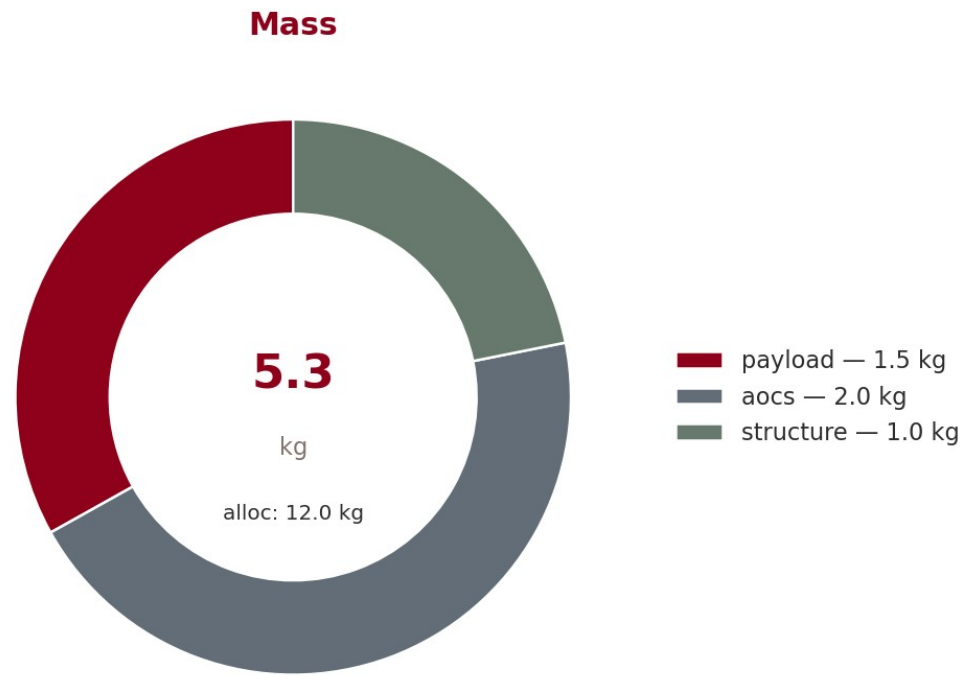


Figure 8 — Mass budget contributions and total (kg).

Subsystem	Equipment	Nominal (kg)	Margin (%)	With margin (kg)
payload	Payload Multispectral Imager Mass	1.5	10	1.65
aocs	AOCS Mass	2.04	20	2.45
structure	Structure Mass	0.99	20	1.19
TOTAL		4.53	56	5.29

Allocation: 12.00 kg — margin 55.9% — status GREEN

13.2 Power budget

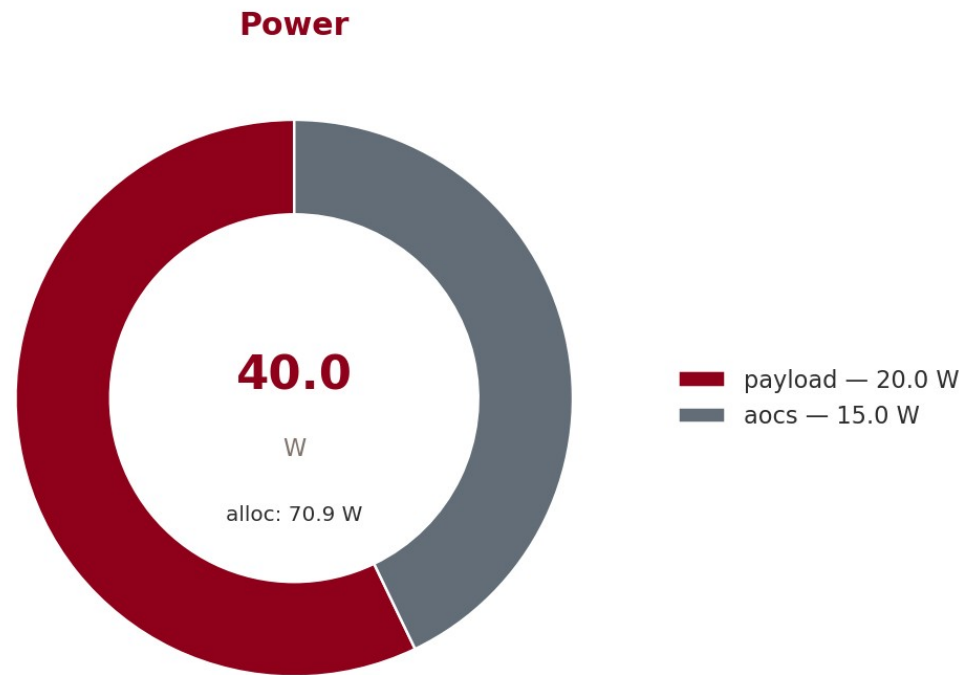


Figure 9 — Power budget contributions and total (W).

Subsystem	Equipment	Nominal (W)	Margin (%)	With margin (W)
payload	Payload Multispectral Imager Power	20	10	22
aocs	AOCS Power	15	20	18
TOTAL		35	44	40

Allocation: 70.90 W — margin 43.6% — status GREEN

13.4 Cost budget

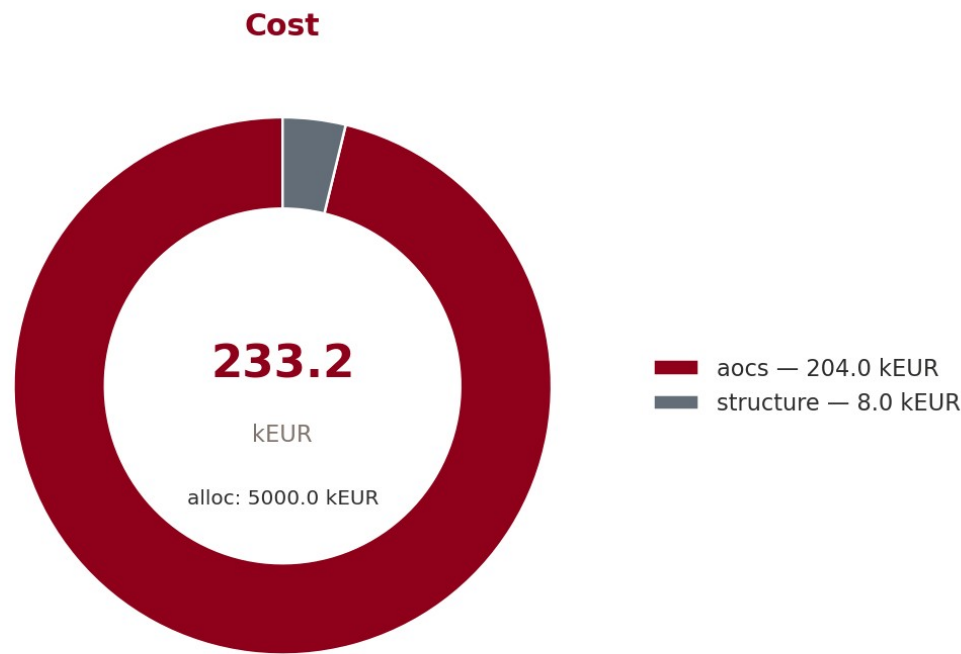


Figure 10 — Cost budget contributions and total (kEUR).

Subsystem	Equipment	Nominal (kEUR)	Margin (%)	With margin (kEUR)
aocs	AOCS Cost	204	10	224
propulsion	Propulsion Cost	0	10	0
structure	Structure Cost	8	10	8.8
TOTAL		212	95	233

Allocation: 5.00 MEUR — margin 95.3% — status GREEN

14. Equipment List

No KB-selected equipment recorded in the design state.

15. Compliance & Verification

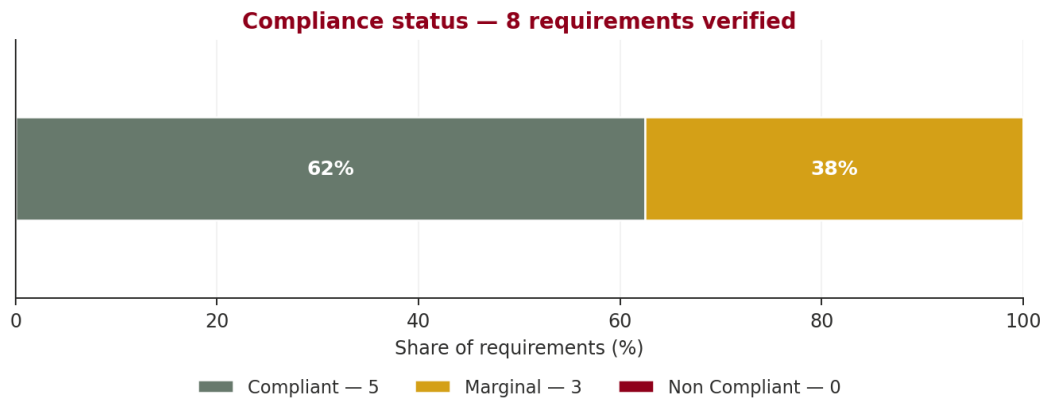


Figure 11 — Compliance status across all requirements.

Req ID	Description	Achieved	Margin %	Status
REQ-SYS-001	The spacecraft shall have a design lifetime of at least 3.0 years	3.00	0.0	marginal
REQ-SYS-002	The spacecraft wet mass shall not exceed 12.0 kg	7.03	41.4	compliant
REQ-SYS-003	The total mission cost shall not exceed 5.0 MEUR	0.70	86.0	compliant
REQ-AOCS-001	The spacecraft shall provide pointing accuracy of 0.1° or better for Multispectral Imager	0.10	0.0	marginal
REQ-DATA-001	The communication system shall support downlink of Multispectral Imager data at 100.0 Mbps	2.00	20.0	marginal
REQ-PWR-001	The EPS shall provide positive power margin in all operating modes including eclipse	70.90	100.0	compliant
REQ-TTC-001	The communication link shall close with at least 3 dB margin at minimum elevation	7.70	156.7	compliant
REQ-SYS-004	The spacecraft shall be capable of deorbiting within 25 years of end-of-mission per space	71.69	100.0	compliant

Total 8 requirements: 5 compliant • 3 marginal • 0 non-compliant (62.5%).

16. Technology Readiness

Subsystem	Baseline (TRL)	Innovation (TRL)	Recommendation
eps	Triple-junction GaAs (Azur Space 3G30C) (9)	Perovskite-silicon tandem solar cells (5)	baseline
eps	Li-ion battery (SAFT VES16) (9)	Solid-state lithium battery (4)	carry_both
aocs	Reaction wheels + star	MEMS gyroscope + AI-based	carry_both

	tracker (9)	attitude determination (5)	
ttc	X-band transponder (conventional) (9)	Optical inter-satellite link (OISL) terminal (6)	baseline
propulsion	Monoprop hydrazine (9)	Water electrolysis green propulsion (5)	baseline
propulsion	Hall-effect thruster (8)	Electrospray (ionic liquid) thruster (6)	carry_both
structure	Aluminium honeycomb panels (9)	3D-printed titanium lattice structure (6)	carry_both
thermal	MLI + radiator panels (9)	Phase-change material thermal buffer (5)	baseline
obdh	RAD750 / GR740 radiation-hardened processor (9)	COTS RISC-V with radiation-tolerant design (5)	baseline

17. Reliability & FMECA

Rationale

Series reliability model $R_{sys} = \prod R_i$. Over 3.0 yr (26298 h) the mission reliability is 0.718 against a target of 0.90. Weakest contributor: aocs (R=0.924). System MTBF is 60241 h.

Assumptions

- Constant failure-rate (exponential) model $R(t) = e^{-\lambda t}$.
- Series model: all subsystems must function; no parallel redundancy modelled here.
- Failure rates from heritage (SMAD4 Table 19-5, MIL-HDBK-217F).

Warnings & open issues

- Mission reliability 0.718 < target 0.90; weakest: aocs (R=0.924). Consider redundancy.
- Single-point failures: eps (R=0.936), aocs (R=0.924), obdh (R=0.949), payload (R=0.924)

17.1 Per-subsystem failure rates

Subsystem	λ (1/h)	R(mission)	Redundancy
EPS	2.50e-06	0.9364	single string
AOCS	3.00e-06	0.9241	single string
TTC	1.50e-06	0.9613	single string
OBDH	2.00e-06	0.9488	single string
TCS	5.00e-07	0.9869	single string
PROPULSION	4.00e-06	1.0000	single string
STRUCTURE	1.00e-07	0.9974	single string
PAYLOAD	3.00e-06	0.9241	single string

17.2 FMECA (top items)

Item	Failure mode	Effect	S	O	D	RPN	Mitigation
EPS	Battery cell short	Loss of bus voltage; depleted battery	4	3	2	24	Cold/hot redundancy; safe-mode handoff; ground-commanded res
AOCS	Reaction-wheel bearing seizure	Loss of fine pointing; switch to coarse mode	4	3	2	24	Cold/hot redundancy; safe-mode handoff; ground-commanded res
OBDH	OBC SEU latch-up	Reset cycle; loss of telemetry window	4	3	2	24	Cold/hot redundancy; safe-mode handoff; ground-commanded res
PAYLOAD	Detector dark current rise	Image SNR degradation; calibration drift	4	3	2	24	Cold/hot redundancy; safe-mode handoff; ground-commanded res
TTC	Transmitter PA failure	Loss of downlink; use backup transmitter	3	2	2	12	Cold/hot redundancy; safe-mode handoff; ground-commanded res
TCS	Heater string open	Component temperature excursion below limit	3	2	2	12	Cold/hot redundancy; safe-mode handoff; ground-commanded res
STRUCTURE	Bolt back-out	Loss of deployable, latched structure	3	2	2	12	Cold/hot redundancy; safe-mode handoff; ground-commanded res

18. Risk Register

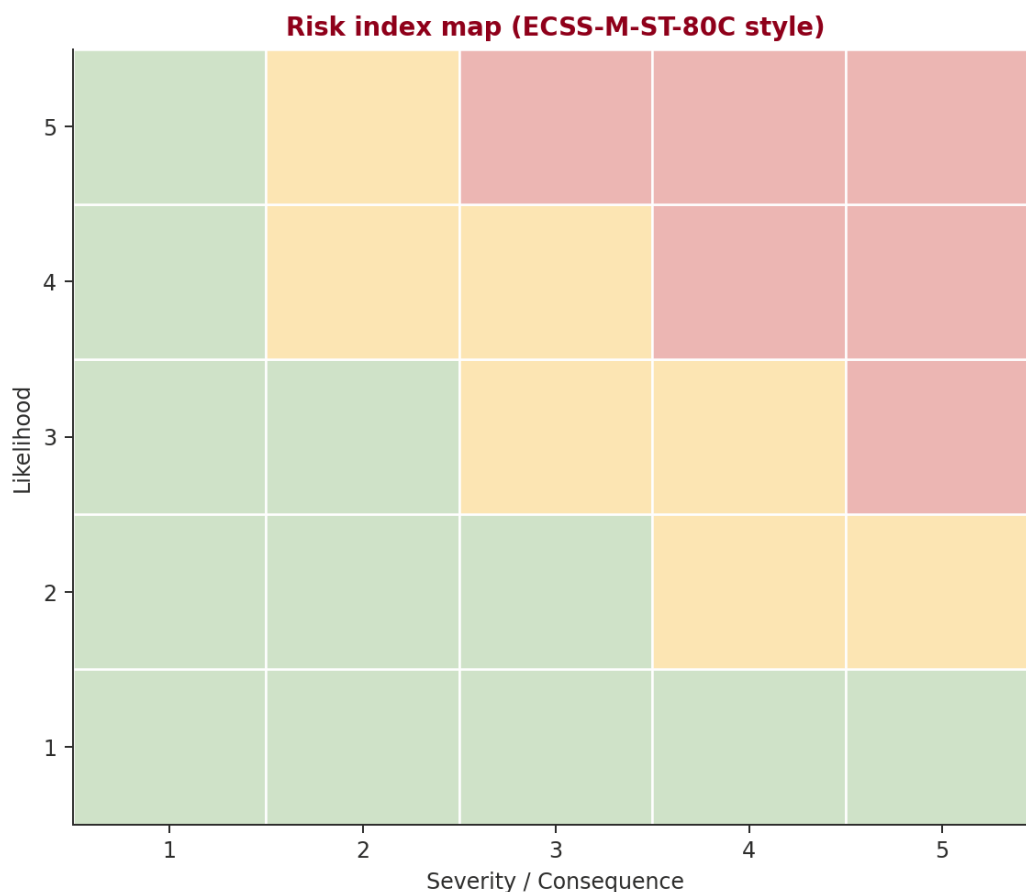


Figure 12 — 5×5 risk index map (ECSS-M-ST-80C).

Risk register pending — no risks identified at this stage.

19. Space Debris Compliance

Rationale

Post-mission disposal: orbital lifetime 2.7 yr with deorbit method 'natural'. Casualty risk $E_c = 1.10 \times 10^{-8}$ (compliant against 10^{-4} threshold). Passivation score 1.00 covers stored-energy depletion of battery and propellant tanks per ISO 24113. 25-yr rule satisfied; 5-yr rule satisfied.

Assumptions

- Atmospheric drag lifetime per NRLMSISE-00 / JR-1971 (mean solar flux).
- Casualty risk per NASA-STD-8719.14 demise model.
- Passivation score weighted by battery + propellant tank stored energy.
- Collision-avoidance ΔV budget assumed ~ 1 m/s per year in LEO.

Parameter	Value
Orbital lifetime (yr)	2.7

25-yr rule (IADC)	Compliant
5-yr rule (FCC)	Compliant
Casualty risk Ec	1.10e-08
Casualty risk compliance	Compliant
Passivation score	1.00
Compliance score (0–100)	92
Deorbit method	natural
CA ΔV per year (m/s)	2.5

20. Cost & Schedule

Rationale

Parametric life-cycle cost estimate using the CubeSat COTS model. Hardware 151 kEUR + AIT 18 kEUR + NRE 45 kEUR (30% of HW) + launch 200 kEUR + operations 300 kEUR over 3.0 yr. Total 0.7 MEUR with ±40% confidence interval at Phase A.

Assumptions

- Spacecraft class 'nano' → COTS pricing table.
- AIT 12% of HW; NRE 30% of HW; ops €100k/yr.
- Launch cost from rideshare/dedicated-launcher market reference.
- All costs in 2025-EUR; no inflation escalation applied.
- Phase-A P-curve: P50≈total, P70≈+15%, P80≈+25%, P90≈+50% per ECSS-M-ST-60C cost-uncertainty practice.

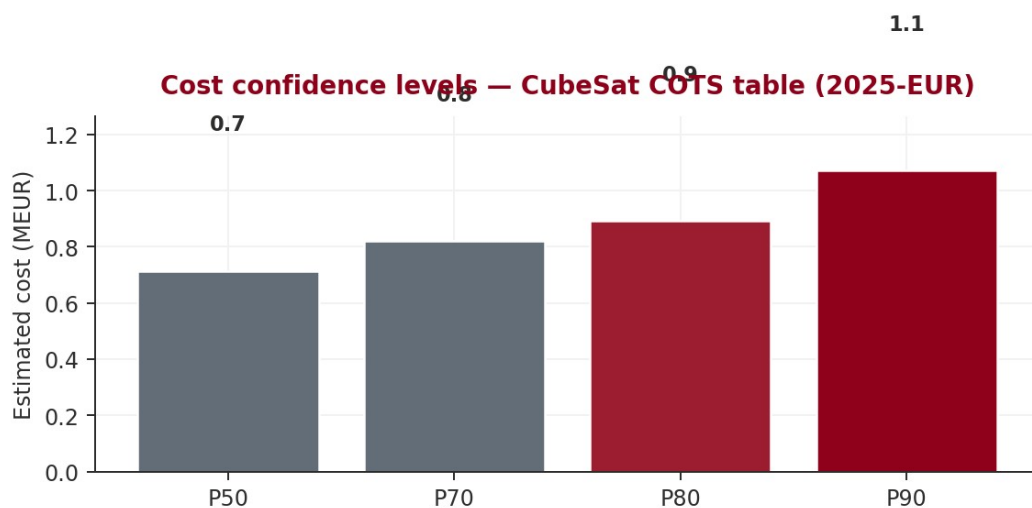


Figure 13 — Cost estimate at P50/P70/P80/P90 confidence levels.

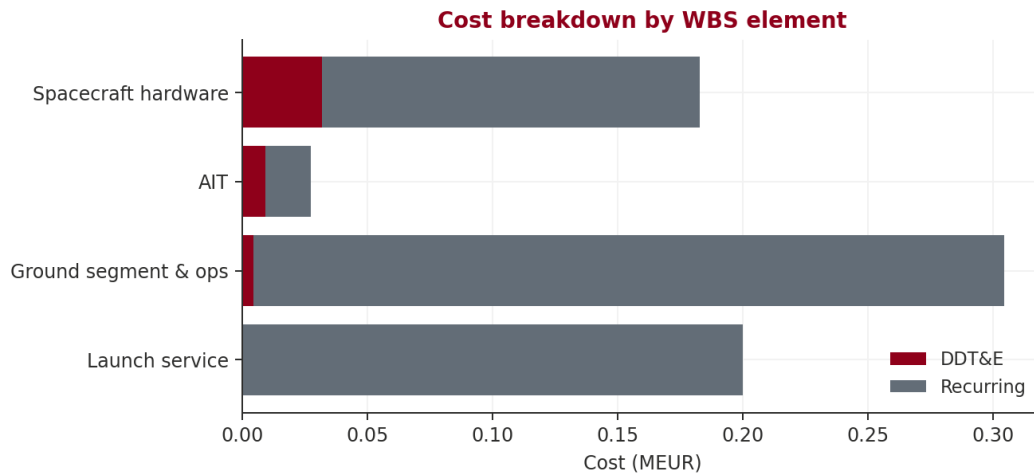


Figure 14 — Cost breakdown by WBS element (DDT&E + recurring).

WBS	Element	DDT&E (MEUR)	Recurring (MEUR)	Total (MEUR)
1.1	Spacecraft hardware	0.03	0.15	0.18
1.2	AIT	0.01	0.02	0.03
1.3	Ground segment & ops	0.00	0.30	0.30
1.4	Launch service	0.00	0.20	0.20

21. Design Loop Convergence

Metric	Value
Converged	Yes
Iterations	16
Wall-clock time (s)	0.02

22. Conclusions & Recommendations

The SRR baseline is now established. Key engineering observations follow.

22.1 Open warnings

- Link margin $-2.6 \text{ dB} < 3 \text{ dB}$ minimum (required rate 100.0 Mbps vs max 54.4 Mbps)
- Radiator area 0.11 m^2 exceeds 50% of S/C area — consider active cooling
- Mission reliability $0.718 < \text{target } 0.90$; weakest: aocs ($R=0.924$). Consider redundancy.
- Single-point failures: eps ($R=0.936$), aocs ($R=0.924$), obdh ($R=0.949$), payload ($R=0.924$)

22.2 Recommendations

- [INNOVATION] STRUCTURE: Consider 3D-printed titanium lattice structure (TRL 6) as tech demo alongside baseline Aluminium honeycomb panels. Benefit: 30% mass reduction, integrated thermal paths. Risk: Qualification for flight loads ongoing
- [INNOVATION] PROPULSION: Consider Electrospray (ionic liquid) thruster (TRL 6) as tech demo alongside baseline Hall-effect thruster. Benefit: Ultra-precise thrust, no moving parts. Risk: Limited total impulse for large dV

22.3 Next steps

- Lock the Mission Requirements Document and Concept of Operations.
- Prepare the system architecture trade space for the PDR.
- Mature low-TRL components and update the technology development plan.